

Inheritance Language: Allele to Phenotype

I can use inheritance vocabulary to connect alleles, genotype and phenotype in a simple monohybrid model.

ALLELE

version of a gene

GENOTYPE

allele combination

PHENOTYPE

observed characteristic

[Open the original interactive lesson](#)

What success looks like

- I can distinguish a gene from an allele.
- I can distinguish genotype from phenotype.
- I can identify homozygous, heterozygous, dominant and recessive combinations without value language.

Retrieve and reconnect

- Define gene.
- Describe how DNA and chromosomes are related.
- State what a base sequence can carry.
- Lead-in: predict the difference between genotype and phenotype.

In a simplified pea model, P is a purple-flower allele and p is a white-flower allele. Which item is a genotype?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Pp
- B. purple flowers
- C. the whole pea plant

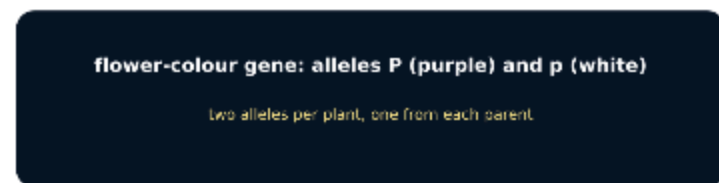
Look closely · what is the evidence?

GENE · DNA section

ALLELE · version of that gene

GENOTYPE → PHENOTYPE in this simplified model

Look closely · what is the evidence?



Gene and alleles



Pp genotype



pp genotype

Words we will use

- allele — version of a gene
- genotype — allele-letter combination
- phenotype — observed characteristic

Watch the model

- Use one simplified pea-flower gene with two alleles: P and p. Capital and lowercase are symbols, not ratings.
- Build genotypes PP, Pp and pp; label PP/pp homozygous and Pp heterozygous.
- Apply complete dominance in this model: PP and Pp give purple phenotype; pp gives white phenotype.

Which statement is correct for the simplified P/p model?

A. Pp is a heterozygous genotype with the purple phenotype.

B. P is dominant because purple is biologically better.

C. pp is heterozygous because the letters are lowercase.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Dominant describes expression in a heterozygote; it does not mean stronger, healthier, better or more common.
- Phenotype is not always determined by one gene alone; this pea model deliberately uses a simple monohybrid characteristic.
- Each gamete carries one allele for this gene, which becomes the starting rule for a Punnett square.

Find and repair the misconception

A dominant allele is stronger, better and always more common.

A. Keep it: dominance is a value ranking.

B. Repair it: dominant describes expression in a heterozygote, not quality or frequency.

C. Repair it: capital letters make alleles physically larger.

Genotype-phenotype decoder

Classify each card, then justify the phenotype using the simplified dominance rule.

- ALLELES
- GENOTYPE
- ZYGOSITY
- PHENOTYPE
- LIMIT

Genotype-phenotype decoder

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- HOMOZYGOUS
- HETEROZYGOUS
- PHENOTYPE STATEMENT

Use genotype, phenotype and homozygous/heterozygous in one accurate explanation.

Your independent mission

Decode six genotypes, identify phenotypes and explain the limits of the simple P/p model.

Record: Accurate use of allele, genotype, phenotype, homozygous and heterozygous plus one model limitation.

Choose your support and challenge

- Supported: Match PP, Pp and pp to same/different alleles and purple/white phenotype.
- Standard: Classify six genotype/phenotype statements and explain two matches.
- Stretch: Correct three dominance misconceptions and evaluate the one-gene complete-dominance model.

Show what you know

Explain why Pp is heterozygous and gives purple flowers in this model.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.